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Roll No

CS-8302 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Digital Image Processing
(Elective - III)
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. With a brief note on various elements of digital image processing system, explain image acquisition devices in detail.
2. a) Discuss about error free compression and lossy compression schemes.
b) How to detect discontinuities by point detection, line detection and edge detection method?
3. Find discrete cosine transform and its inverse for the following image data:
[0 255; 255 0] [2×2] matrix.
4. Develop an algorithm for Fast Hadamard transform.
5. For $V = \{1\}$, determine whether the following pixels are
 - i) 8 adjacent
 - ii) m adjacent
$$\begin{matrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$
6. Explain the following operations:
 - i) Erosion
 - ii) Dilation
 - iii) Operation
 - iv) Cloning
 - v) Thinning
7. a) Discuss edge detection. Show prewitt and sobel mask applications for diagonal edge detection.
b) Explain Global and adaptive thresholding techniques.
OR
 - a) Write an algorithm to estimate thresholding values.
 - b) Explain JPEG baseline coding and decoding with the help of suitable example.
8. Write short notes:(Any two)
 - a) Morphology
 - b) Signatures
 - c) Polynomial Approximation
